

Neutral-Atom Quantum Optimization

1. Pasqal tech: the platform and how optimization maps onto it

Quantum computing with neutral atoms

Henriet, Beguin, Signoles, Lahaye, Browaeys, Reymond, Jurczak | *Quantum* 4, 327 (2020)

[arXiv:2006.12326](#)

The foundational paper. It reviews the devices from atoms and qubits up to application interfaces, and classifies what is already addressable in the NISQ era, covering both digital mode (gate-based circuits) and analog mode (Hamiltonian sequences).

Graph algorithms with neutral atom quantum processors

Dalyac, Leclerc, Vignoli, Djellabi, da Silva Coelho, Ximenez, Dareau, Dreon, Elfving, Signoles, Henry, Henriet | *EPJ A* 60, 177 (2024)

[arXiv:2403.11931](#)

The best single overview of the optimization side of the platform. Read it straight after the 2020 review.

Efficient protocol for solving combinatorial graph problems on neutral-atom quantum processors

da Silva Coelho, D'Arcangelo, Henry | 2022

[arXiv:2207.13030](#)

The encoding and embedding mechanics that sit underneath everything which follows.

Identifying hard native instances for the maximum independent set problem on neutral atoms quantum processors

Cazals, François, Henriet, Leclerc, Marin, Naghmouchi, da Silva Coelho, Sikora, Vitale, Watrigant, Witt Garzillo, Dalyac | 2025

[arXiv:2502.04291](#)

Builds MIS instances that are embeddable on the QPU and at the same time genuinely hard classically, since earlier benchmark instances were structurally too easy. It concludes that scaling to around a thousand atoms at a 1 kHz repetition rate is a necessary step toward computational advantage. Share this one deliberately: it sets honest expectations and earns credibility with technical readers.

Quantum optimization on Rydberg atom arrays with arbitrary connectivity: gadgets limitations and a heuristic approach

Cazals, Sorondo, Onofre, Dalyac, da Silva Coelho, Vitale | 2025

[arXiv:2508.06130](#)

Answers the question every newcomer asks: what about graphs that are not unit-disk? It proves that any polynomial reduction to unit-disk instances costs a quadratic blow-up in vertex count and degrades approximation guarantees, then offers a divide-and-conquer heuristic with only linear overhead, demonstrated on Orion Alpha. That heuristic is reused in the molecular docking paper in section 3.

2. QUBOs: the modelling language

Field context rather than Pasqal research, but the vocabulary needed for sections 3 and 4.

A tutorial on formulating and using QUBO models (Quantum Bridge Analytics I)

Glover, Kochenberger, Du | 2019

[arXiv:1811.11538](#)

The canonical starting point. It shows how the QUBO model unifies a wide range of combinatorial optimization problems, and how its equivalence to the Ising model connects it to quantum hardware. If someone reads only one QUBO reference, this is it.

Applications and computational advances for solving the QUBO model

Glover, Kochenberger, Du | Chapter 2, pp. 39–56, in Punnen (ed.), *The Quadratic Unconstrained Binary Optimization Problem*, Springer 2022

[Chapter PDF](#) · [Publisher](#)

The applications companion to the tutorial: concrete formulations and computational results across problems found in industry and government.

What works best when? A systematic evaluation of heuristics for Max-Cut and QUBO

Dunning, Gupta, Silberholz | *INFORMS Journal on Computing* 30(3), 608–624 (2018)

[doi:10.1287/ijoc.2017.0798](https://doi.org/10.1287/ijoc.2017.0798)

The classical baseline. No single heuristic outperforms the others across all instances, which is exactly the framing needed before anyone asks whether a QPU beats classical solvers.

Optimized QUBO formulation methods for quantum computing

De Santis, Tirone, Marmi, Giovannetti | *Quantum Science and Technology* (2025)

[Publisher](#) · [arXiv:2406.07681](https://arxiv.org/abs/2406.07681)

How to cut the variable count in a reformulation: efficient use of slack variables even for problems with non-linear constraints, via the iterative quadratic polynomial and master-satellite methods. Directly relevant to fitting real problems into a limited qubit budget.

Towards arbitrary QUBO optimization: analysis of classical and quantum-activated feedforward neural networks

Lai, Blank, Schmelcher, Mukherjee | *Machine Learning: Science and Technology* (2025)

[Publisher](#) · [arXiv:2410.12636](https://arxiv.org/abs/2410.12636)

A strong purely classical neural QUBO solver, reported as outperforming Gurobi by 72 percent on 200-variable random QUBOs within a 100-second limit, then extended with a quantum-annealer-based activation function into a quantum-classical encoder-decoder. A useful reminder of how high the classical bar sits.

3. Applications

A scalable heuristic for molecular docking on neutral-atom quantum processors

Garrigues, Onofre, Coelho, Acheche (Pasqal) | 2025

[arXiv:2508.18147](https://arxiv.org/abs/2508.18147)

Docking mapped to a Maximum Weighted Independent Set problem, with the size mismatch between biologically relevant systems and near-term devices broken by the divide-and-conquer heuristic from Cazals et al. Read it right after [arXiv:2508.06130](https://arxiv.org/abs/2508.06130).

Satellite mission planning with Rydberg atoms

Nowak, Marchand, Naghmouchi, Rainjonneau, Coelho, Vignoli, Henry (Thales cortAix and Thales Alenia Space with Pasqal) | 2026

[arXiv:2606.23045](https://arxiv.org/abs/2606.23045)

Daily scheduling for an Earth-observation satellite fleet against a dense set of targets, converted step by step into a cold-atom-friendly formulation: first naively, then accounting for the agility of the satellites. A good customer-facing example of the modelling journey, not just the final result.

4. More complex, hybrid approaches

The recurring pattern here: the QPU is used as a sampler inside an established classical operations-research framework.

Quantum pricing-based column-generation framework for hard combinatorial problems

da Silva Coelho, Henriët, Henry | *Phys. Rev. A* 107, 032426 (2023)

[arXiv:2301.02637](https://arxiv.org/abs/2301.02637)

The template for this whole section, benchmarked on the Minimum Vertex Coloring problem. Start here.

Hybrid quantum-classical branch-and-price method for the vertex coloring problem

Vercellino, Naghmouchi, Coelho, Vitali, Scionti, Viviani, Terzo, Montrucchio (Fondazione LINKS and Politecnico di Torino with Pasqal) | 2025

[arXiv:2508.18887](#)

The natural sequel: quantum-assisted column generation pushed up into a full branch-and-price scheme, reaching optimality on roughly 98 percent of benchmark instances, with preliminary validation on the Orion Alpha QPU showing robustness to noise and hardware constraints.

Mixed integer linear programming solver using Benders decomposition assisted by neutral atom quantum processor

Naghmouchi, da Silva Coelho | *Phys. Rev. A* 110, 012434 (2024)

[arXiv:2402.05748](#)

A different classical decomposition. The MILP master problem is converted to a QUBO model by an automated procedure and solved on the device, with the conversion tightening the bounds of the continuous variables to reduce the required qubit count.

Enhancing the performance of quantum neutral-atom-assisted Benders decomposition

Joliot, Naghmouchi, Coelho (Pasqal) | 2025

[arXiv:2503.03518](#)

The follow-up, and worth reading as a pair with the previous entry. It adds robust feasibility cuts, slack-free exponential encoding, constructive penalty tuning and a multi-cut strategy, taking the feasibility rate from 68 to 100 percent and the optimality rate from 52 to 86 percent. It is also candid about where the earlier proof-of-concept fell short, which makes it an unusually instructive read.